Hack-Nation

In collaboration with MIT Club of Northern California and MIT Club of Germany

CHALLENGE

04

SMALL AI FOR DEVELOPMENT HACKATHON

What does localizing AI development mean for you?

Concept Note · Youth Summit × Global AI & Digital Summit (Seoul, October 2026)

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The World Bank – Youth Summit

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01 Background

As of 2024, 2.6 billion people—about one-third of the world's population—remain offline, and internet use ranges from over 90% in high-income countries to just 27% in low-income countries. Meanwhile, 800 million people still lack official ID, limiting access to digital public services, finance, and healthcare. Closing that divide is the mandate of the World Bank Group's Digital & AI Vice Presidency, which supports countries to expand affordable, inclusive broadband connectivity, bring people onto digitally enabled services, and bring more women online, commitments tracked on the World Bank Group Scorecard.

Employment is the second challenge. Over the next decade, an estimated 1.2 billion young people in the Global South will reach working age, while current trends are projected to generate only 420 million jobs.

Artificial intelligence and the broader data revolution are expanding what institutions can do, improving service delivery, and opening new economic pathways. Yet not everyone is benefiting equally. This challenge sits inside that agenda. Most AI applications have been deployed and designed with the assumption of baseline access to reliable connectivity, use of a smart-model device, and a user already familiar with how to prompt when interacting with an AI interface. This challenge is designed for the many settings where these conditions are either strained, limited, or intermittent.

02 What is Small AI

Small AI is a design approach for AI development that promotes targeted, use-case-driven AI: eg. while general-purpose large language models continue to improve their viability in constrained environments, a *Small AI* approach focuses on leveraging AI capabilities — pattern recognition, text generation, or the ability to understand context[1] — to work on a single, well-defined problem, on a device easily accessible to users. That device may be a basic phone, or a small piece of on-device hardware such as a sensor or a microcontroller that runs the model locally. The AI use case may be predictive, generative, or agentic; what matters is that its targeted use, less reliance on the existence of an extensive data-center infrastructure, functions where connectivity is weak or absent, can be trained or tuned on local data, and centers the user through common interfaces such as SMS and voice. The design is governed by the operating constraint and the problem to be solved, not by the scale or capability of the model.

This comes with an important caveat. A good AI solution is not sufficient on its own. Small AI depends on a set of foundational enablers — adequate connectivity, digital literacy, and trusted local institutions — and without them even the best tool remains a pilot. Each sector brief below therefore sets out these preconditions alongside the challenges specific to the sector, so that a proposed AI solution can demonstrate a clear and distinct value add. Applicants should address **one sectoral challenge** to build their Small AI solution.

[1] Vianney Gomezgil Yaspik, Digital and Computational Studies Assistant Professor of Digital and Computational Studies

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03 What is the Small AI for Development Hackathon?

In this regard, the World Bank Group Digital & AI Vice Presidency, in collaboration with the Youth Summit, Development Economics Vice Presidency (DEC), Information and Technology Solutions (ITS) and Hack-nation, is organizing a hackathon to bring the youth perspective into how AI is developed. Building on the World Bank Group's work to help countries adopt and scale AI, including the AI Repository, a catalogue of AI use cases and the lessons from deploying them. Over one weekend, applicants will build a working Small AI solution to one of the sector challenges below.

Through their Small AI solution, applicants should seek to answer: **what does localizing AI development mean for you?** Young people are often closest to the challenges this competition addresses, and their perspective on how technology should respond is essential. This challenge invites young people to consider what it means to develop AI for their own context, language, and local needs, and to reflect candidly on both the opportunities and the risks. Entries that acknowledge these trade-offs are encouraged.

04 Eligibility, Timeline and Logistics

The competition is open to entrants aged 18 to 35, in line with the United Nations' definition of youth, open to participants from over 150 countries. Applicants from emerging economies are encouraged to apply.

Key dates

- Application Opens: September 2026
- Competition weekend: 3–4 October 2026
- Shortlist selection by expert evaluators: 5–6 October
- Ignite Talk by the winners in Seoul: 21 October 2026

Entries submitted during the competition weekend will be reviewed by expert evaluators drawn from health, agriculture, and tourism, who will select the shortlist for final judging. Each sector panel selects a shortlist of finalists in its own sector, and one winning team is chosen per sector, three winners in total.

One representative from each of the three winning teams, one from each sector, will be invited to Seoul to present at the Global AI and Digital Summit 2026, co-hosted by the World Bank Group and the Government of Korea from 19 to 22 October. The Summit convenes governments, the private sector, investors, and international organizations to examine how practical, fit-for-purpose AI can create jobs and expand opportunities in developing economies, with dedicated attention to health, agriculture, tourism, education, and government services. The Ignite Talks will take place in a breakout session. Organizers will facilitate networking opportunities and, where feasible, support introductions to investors, private-sector representatives, MSMEs, and government officials attending the event. Travel expenses will be covered. Visa arrangements will be communicated to each winning team directly (one representative per team will be traveling).

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In summary

WHAT IT IS	A Small AI for Development hackathon challenge for young innovators, held jointly by the World Bank Group Youth Summit and the Global AI & Digital Summit in Seoul.
WHO CAN ENTER	Young innovators aged 18 to 35 (the UN definition of youth).
PARTNER	Hack-Nation will manage the competition and submit a shortlist of finalists to the Bank for final judging
HOW IT RUNS	Application and Promotion through Hack-Nation opens in September, a competition weekend in early October, judging by World Bank Group Bank panel in each sector, then a showcase in Seoul.
KEY DATES	Competition Oct 3 to 4; shortlists on Oct 5-6; the winning teams travel by Oct 18; Ignite Talks on Oct 21.
WHAT THE WINNER GETS	One winning team per sector, three in total. One representative per team receives flight and lodging to Seoul, a spotlight session at the summit, and opportunities to engage with government, private-sector, and development leaders.

05 The Challenge

Who you are building for

Over one weekend, you build one Small AI tool/solution for one sector and demonstrate it works within the constraints specified within each sector. Each sector's scenario, limits, evidence, and datasets are in its annex. (*Annex A: Health, Annex B: Agriculture, Annex C: Tourism*)

Meet Noor

Noor is 38, farms 2 hectares in the Ondera highlands; coffee on the upper slope, maize and beans below and has been a member of the Ondera Coffee Cooperative for eleven years. Her 16-year-old daughter boards at school in the district town. At home she speaks her local language, and the national one when she needs it like most of her neighbors do when they have to interact with tourists every now and then. Two phones serve the household: her own, which she uses for calls, messages, and mobile money, and her daughter's smartphone, which she only really uses when her daughter is home on weekends to set it up and show her how, most often to watch the news on YouTube. There is no Wi-Fi at the house; they buy 3G data bundles from the local network when they need them. For most of the day she is out on the slope, and the phone is at the house.

Noor is fictional; her constraints are not. They are drawn from World Bank-supported work with governments, including primary health care access in Solomon Islands, farmers' access

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to digital advisory and markets in Cote d'Ivoire and tourism and small operators in The Gambia.

Choose your sector

01 Health

She has a clinic near where she lives but it is overcrowded and the quality of care is not always of a good standard. Doctors are well-meaning but are not always up to date with the latest medical guidance. The significant patient load, coupled with burdensome record-keeping requirements, means that they are not able to give each patient the attention that is needed.

Full brief and datasets: ***Annex A****: the scenario, the limits, the evidence, and the extended health dataset list.*

02 Agriculture

Her coffee yields have slipped and she is not sure why. The extension officer reached her village twice last year, and at harvest a buyer names a price she has no independent reference for.

Full brief and datasets: ***Annex B****: the scenario, the limits, the evidence, and the extended agriculture dataset list.*

03 Tourism *turning interest into a business*

Six or seven visitors a month find her farm by word of mouth. Noor needs to turn informal visitor interest into a manageable, trusted customer workflow, be able to communicate with customers who don't speak her language and leverage good reviews she's getting to upgrade her farm tours.

Full brief and datasets: ***Annex C****: the scenario, the limits, the evidence, and the extended tourism dataset list.*

Read your sector's annex before building. It gives the constraints and evidence; the design choices are yours to make and justify.

What you will build

Pick one of the three sector challenges above (Health, Agriculture, or Tourism) and **build an AI-powered tool that addresses it**. By the end of the weekend, you should have:

- **A working prototype:** a phone app, a chatbot, an SMS service, a voice line, etc., that addresses one or more of Noor's challenges. No-code and vibe-coded builds welcome.
- **A clear answer to what the AI-powered tool** (i.e. machine learning, natural language processing, computer vision, generative AI, etc.) is in use and how it is

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creating value or better use of technology compared to other digital tools. If SMS, a spreadsheet, or a Google search could do the same job, AI may not be the best tool to invest in.

- **Proof it works on at least one sector**: an image it helps classify, a conversation it helps translate, a set of reviews helps analyze? Be creative.

06 The Rules

Be creative, think out of the box, whatever form your solution takes:

- it runs on a device the user already has;
- its core feature works offline;
- its model files are small enough to side-load or send over a weak connection;
- at least one interaction is in a local language, by voice or text — name the language, and expect to be asked how the tool would fare in a less-supported one.

> **AI Guardrails**
>
> — Human-in-the-loop principles: A person makes the final call. The tool informs a decision and flags what it is unsure of; it does not act on the user's behalf. If an entry proposes an agentic workflow, ensure the AI checks in with user appropriately.
>
> — Avoid hallucinations.

07 The Data

Every entry should cite its data sources. The provided datasets are just suggestions, not a required list: you are encouraged to find better. Check the terms yourself before building on anything here, as access conditions change.

7.1 How the Datasets are organized

The datasets are arranged in two layers, and a strong entry uses both.

Where to look	What is in it
Common datasets *Section 7.3*	Open to every entry, whichever sector you choose. Four groups: A language and speech; B connectivity, devices and inclusion; C maps, population and satellite; D country statistics and household surveys. They are listed once, there, rather than repeated in each annex.
Sector datasets *Your annex*	Specific to one sector — health in Annex A, agriculture in Annex B, tourism in Annex C. You only need the annex for the sector you chose.

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In practice you will draw on both. The common tables give you the language your tool speaks, the connectivity it has to survive, and the evidence that the problem is real. Your sector table gives you the thing your tool actually works on. Each annex ends by naming the two or three common datasets that matter most for that sector, so you do not have to read all four groups to find them.

7.2 The two kinds of data

1. Data that shows the problem and gap being filled by the solution. Surface data that puts Noor's situation in context — health-worker absence rates, travel time to the nearest facility, the gap between phone and smartphone ownership among women. Cite source, year, and country. Make note if the closest figures come from synthetic data modeling an article or published journal

2. Data you build with. What your tool learns from: speech recordings, labeled photographs, example messages. Name every dataset, its source, its license and its size. **You must also indicate what your data does not cover**, and this is scored. *Most crop-disease datasets, for instance, are studio images on plain backgrounds; a model trained only on them performs poorly on real field photos*. You may supplement with synthetic data as long as you label it accordingly.

7.3 Common Datasets for Every Sector

A. Language and speech

The starting point for any tool that has to listen, speak, translate or understand in a local language.

FOR VOICE AND LANGUAGE		
Dataset	**What it is**	**Why it matters here**
Mozilla Common Voice	Crowdsourced voice recordings, CC0, covering many African and South Asian languages.	The main starting point for listening in a local language. It accepts contributions, so your own recordings can be given back.
FLEURS (Google)	Speech benchmark across roughly 100 languages.	Lets you compare results against a published standard rather than asserting them.
MMS (Meta)	Speech recognition and synthesis covering over a thousand languages.	Often the only usable starting point for a language with little data.
FLORES-200 / NLLB-200 (Meta)	Translation benchmark and open models across 200 languages.	For any interaction where the message arrives in a language your user cannot read.
OPUS	A large open collection of parallel corpora, meaning the same text in two languages side by side.	The raw material for translation between low-resource languages

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		when no ready-made model exists.
MASSIVE (Amazon)	About one million short utterances in 51 languages, labelled by intent.	Sorting incoming messages or requests into a fixed list of intents.
Masakhane	African-language NLP resources built by an African research community.	On-theme as well as useful: localized AI built by local people.
AI4Bharat / IndicVoices	Speech and language resources for Indian languages.	The equivalent starting point for South Asian entries.

B. Connectivity, Devices, and Inclusion

What device your user actually has, whether there is a signal where she lives, and whether she can be paid

GSMA Mobile Gender Gap Report	Phone and smartphone ownership by gender and country: evidence for the device Noor actually has.
OpenCelliD	Open database of mobile cell towers. Show where there genuinely is no signal rather than assuming it
Global Findex (World Bank)	Financial inclusion, including mobile money use, disaggregated by gender.

C. Maps, Population, and Satellite

Where things are, how many people are there, and what the land is doing. Useful in all three sectors — for offline base maps, catchment size, and land or crop condition over time.

WorldPop	Gridded population estimates, for the size of the catchment your tool serves.
OpenStreetMap	Crowdsourced roads, buildings and points of interest, downloadable for offline use.
VIIRS Nighttime Lights	Monthly night-light raster imagery, widely used as a proxy for electrification and economic activity.

D. Country Statistics and Household Surveys

Where you establish the problem at a country level and with a year attached.

World Bank Data360	The Bank's integrated data catalogue, bringing indicator and survey holdings into one place across all sectors.

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World Development Indicators	More than 1,400 development indicators by country and year.
World Bank Microdata Library and Data Catalog	The institution's own survey and household microdata holdings. Start here for evidence that the problem is real.
Humanitarian Data Exchange	UN OCHA's open platform: administrative boundaries, prices, population, infrastructure.

7.4 Your sector's datasets

The second layer, in your annex: Annex A for health, Annex B for agriculture, Annex C for tourism.

08 The Deliverables: what you will submit

Before the weekend: the application form through Hack-nation's Global AI Hackathon website

[Oct 3-4th 2026]: **By the end of the Hackathon weekend,** make sure to submit:

1. **Your prototype** — the working tool, with the code, or a link to it.
2. **A video, 2 to 5 minutes.** (Entries without this will not make it to the shortlist)
   - **Problem statement** — one sentence: ***Because of this tool, [user] will [action] by [when] that they would otherwise [not do / do late / do worse]; we know because [evidence].***
   - **AI capabilities of the solution** — what the tool does with AI (machine learning, computer vision, language, generative), and why a simpler tool (SMS, a spreadsheet, a search) would not do the same job. Mention the guardrails in place.
   - **Tool demo** — the prototype demo with clear user journey end to end (in form of a slide deck, screen recording showing how the tool you built works)
   - **The challenge/gap being addressed** — where the tool sits in the user's day: when they open it, what they do, what happens next. For more technical builds, please include tech stack details.
   - **Your take** — what localizing AI development means to you.

09 Judging Criteria

Criterion	Weight	The question to ask
The built solution (Small AI fidelity)	**25%**	*Does the tool work end to end within the constraints of the sector*

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Development relevance and impact	**20%**	Is this a real problem from the sector briefs, and does the outcome matter to the person it is built for?
Data grounding	**15%**	Does the tool help address an identified gap in the data, is the data modeling sound?
Evidence it works	**15%**	Does the solution fit the challenges identified in the sector, does it add other constraints?
Clarity, design and inclusivity **Value proposition for AI**	**15%**	What the tool does with AI (machine learning, computer vision, language, generative), and would a simpler tool (SMS, a spreadsheet, a search) do the same job?
Scalability, replicability and what happens next	**10%**	Could another setting reuse this innovation?
Responsible AI, data and safety	**Pass/fail**	Are the limits respected, and the account of privacy, consent, bias, and human oversight credible?

How the pass/fail works

How AI performs and impacts these sectors is still an emerging question, and a confident wrong answer is costly when it touches someone's access to essential services or their livelihood. In this regard, each entry should incorporate a fail-safe: AI signposting to a decision-maker when the data it is acting on is not enough to give a definitive answer eg. "not sure — ask a person" rather than guessing, and a human stays in the loop where relevant.

10 Glossary

TERM · WHAT IT MEANS	
Small AI	**Small AI** is a design approach for AI development that promotes targeted, use-case-driven AI: e.g. while general-purpose large language models continue to improve their viability in constrained environments, a *Small AI* approach focuses on leveraging AI capabilities — pattern recognition, text generation, or the ability to understand context — to work on a single, well-defined problem, on a device easily accessible to users.
Offline / on-device (edge) AI	Running the model on the user's own phone or hardware instead of a server, so it works with no connection
Fine-tuning	Adapting an existing, pre-trained model to your case — your language, your crop — by training it further on a smaller, focused dataset. Faster and cheaper than building one from scratch.
Quantization	Shrinking a model so it fits on a basic device or downloads over a weak link.
Store-and-forward	Save it now, send it later. The record waits on the phone until a signal appears.
Dataset	A collection of examples — recordings, photographs, messages, survey answers — that a model learns from or is tested against.

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Training / fine-tuning	Teaching a model using examples. Fine-tuning means adjusting an existing model for your case rather than building one from scratch.
Benchmark	A published standard test that lets different teams' results be compared.
Fixed list of answers	The complete set of things your tool is allowed to say. If it can say anything, it cannot be checked for safety.
Georeferenced	Each record has real coordinates attached, so it can be mapped.
Speech recognition	Turning spoken words into written text.
Intent classification	Working out what someone wants from what they wrote — a price, directions, a booking.
Computer vision	The use of AI to extract useful information from an image, such as identifying a disease on a coffee leaf.
Low-resource language	A language with little digital text or recorded speech available, which is why most AI tools support it poorly.

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Annex A. Health

A.1 The scenario

Noor has a clinic near where she lives, but it is overcrowded and the quality of care is not always of a good standard. Doctors are well-meaning but are not always up to date with the latest medical guidance. The significant patient load, coupled with burdensome record-keeping requirements, means that they are not able to give each patient the attention that is needed.

- **The problem**. Access to primary care in remote catchments is limited by clinician scarcity and the absence of on-site diagnostics. Patients travel long distances to be seen, health workers have limited time per case, and low patient digital literacy compounds both.
- **What other digital tools have been in use.** Basic mobile technology has delivered measurable health gains so far. The Mwana project in Zambia and Malawi ran entirely on SMS: built on the RapidSMS platform, it transmitted infant HIV test results from central laboratories back to rural clinics and sent automated appointment reminders to mothers, cutting result turnaround times sharply. Uganda's mTrac used the same SMS foundation to collect weekly data from health facilities into the national health information system and to push medicine stock-out alerts, reaching over 10 million subscribers. Both worked through message routing and data aggregation. The Broadband Commission estimated that wider mHealth adoption in Sub-Saharan Africa could save approximately one million lives over five years against malaria, HIV/AIDS, and perinatal conditions.
- **Where AI can add value.** The clinical use of AI in low-resource health with the strongest evidence base is image-based screening. In 2021, the World Health Organization endorsed computer-aided detection (CAD) software for chest X-ray interpretation in tuberculosis screening in individuals aged 15 and older, based on the evaluation of three products — CAD4TB, Lunit INSIGHT CXR, and qXR — that performed comparably to, and in later versions better than, human readers. These tools are now deployed in active case-finding programmes across India, the Philippines, and some countries in Africa, often paired with ultra-portable X-ray hardware. AI-enabled retinal screening for diabetic retinopathy and visual inspection tools for cervical pre-cancer follow the same pattern: the model reads an image and flags the cases that warrant a clinician's review.
- **The risks.** The World Health Organization (WHO)'s *Ethics and governance of artificial intelligence for health* is explicit that opportunities and risks are linked, and cautions against unethical collection of health data, biases encoded in algorithms, and risks to patient safety and cybersecurity. Models trained largely on data from high-income countries can underperform on the populations they are then deployed to. Bias can be encoded silently and reproduced across many patients before it is detected. Patient data moving through a mobile network raises safety exposures that a paper record does not. A confident wrong output where a life is at stake is worse than no output. Applicants should design for these risks from the start.
- **Preconditions to keep in mind.** International Finance Corporation (IFC)'s TechEmerge Health program covered Kenya, Uganda, and Ethiopia. Even its strongest AI-assisted ECG and diagnostic tools needed 2G/3G to work. Clinician trust and regulatory acceptance took years to build before anything scaled. Health

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entries touch patient data. Applicants should state where the data sits, who can read it, and what happens when the phone is lost or shared.

> **Hackathon challenge**
> Design and demonstrate a Small AI solution that improves one meaningful part of Noor's access to primary care or a frontline worker's ability to serve her; for example screening support, documentation, referral, follow-up or continuity of care.

A.2 Datasets, extended

This annex lists the health datasets only — the second of the two layers described in Section 7.1. Every entry can also draw on the common datasets in Section 7.3, which are open to all three sectors.

Note the hard limits above: this sector deliberately lists no medical imaging or diagnosis datasets, because interpreting them is out of bounds.

FOR HEALTH ACCESS		
Dataset	**What it is**	**Why it matters here**
Service Delivery Indicators (World Bank)	Facility surveys: provider absenteeism, staffing, equipment, drug availability.	Literally the "is Tobi there today" problem, measured. The strongest single source for this sector.
healthsites.io	Open, OpenStreetMap-based register of health facility locations worldwide.	Answers "where is the nearest facility" with coordinates.
Maina et al., Scientific Data	About 98,000 geolocated public health facilities in sub-Saharan Africa.	A published facility list.
DHS Program and Service Provision Assessments	Health-seeking behaviour; facility staffing, hours and stock. Free registration.	How far people travel, and what they find on arrival.
Malaria Atlas Project travel-time surfaces	Modelled travel time to the nearest town or health facility.	Useful travel data.
AccessMod (WHO)	Modelling geographic access to health services.	Useful if your solution touches which facility to send someone to.
DHIS2	The open-source health information platform used by ministries of health in more than 70 countries, with demo instances and documentation.	The system your record most plausibly lands in. If your tool feeds a facility record, this is what "institutional fit" looks like.

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WHO Global Health Observatory	Country-level health indicators, workforce density, service coverage.	Problem-is-real evidence: workforce shortages and coverage gaps, cited by country and year.
Global Health Data Exchange (IHME)	Catalogue of health surveys, censuses and burden-of-disease estimates.	A second, citable source for establishing the scale of the access problem.

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Annex B. Agriculture

B1. Scenario

Noor's coffee yields have dropped this season, and she cannot say why. The nearest extension officer visits the sub-county twice a year at best, and at harvest she sells her parchment to whichever middleman drives up the valley, at whatever price he names.

- **The problem.** Smallholder farmers lack timely, localized advice on what is affecting their crop, and they lack independent information on what that crop is worth when it is time to sell. Public extension services are constrained by staff shortages, manual data collection, and delayed alerts. In many settings, the binding constraint is the absence of a working farmer registry rather than the absence of an algorithm; without a registry, farmers cannot be reached at scale by any tool, digital or otherwise.
- **What other digital tools have been in use.** In Côte d'Ivoire, the World Bank has supported digital platforms that connect farmers to advisory content and to markets, registering producers and delivering price and input information through the web and mobile channels. The World Bank Group's AgriConnect initiative aims to reach 300 million farmers and double annual agribusiness investment to $9 billion by 2030.
- **Where AI can add value.** India's Wadhwani AI cotton pest tool uses computer vision: a farmer photographs a pheromone trap, and the model counts and identifies the pests on the image, then advises whether and when to spray. That is pattern recognition across thousands of visual examples, compressed to run on a basic smartphone offline. Voice-based advisory in local languages is a second line of promising work, particularly where literacy or screen literacy is a constraint.
- **Preconditions to keep in mind.** In Ukraine, a working farmer registry unlocked advisory services, insurance, and grants for 150,000 farmers. AI stacked on absent registries, absent phones, or absent trust in the advisory system will be difficult to implement in practice.

> **Hackathon challenge**
>
> Design and demonstrate a Small AI solution that helps Noor make, communicate or act on one better agricultural decision; for example identifying a crop or post-harvest problem, timing a farming activity, accessing localized advisory, documenting a field observation, improving quality, value addition or connecting evidence to a pricing, market or extension-service next step.

B.2 Datasets, extended

This annex lists the agriculture datasets only — the second of the two layers described in Section 7.1. Every entry can also draw on the common datasets in Section 7.3, which are open to all three sectors.

FOR AGRICULTURE		
Dataset	**What it is**	**Why it matters here**

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LSMS-ISA (World Bank Microdata Library)	Georeferenced farm household panels across several African countries.	Hosts smallholder farmer data
Cassava Leaf Disease and iBean (Makerere University)	Crop disease photographs taken in real fields by real farmers.	Field conditions closer to scenario
PlantVillage	About 54,000 leaf images, 38 disease classes.	Plain-background studio images. Use it and be honest about the gap.
PlantDoc	About 2,600 crop disease images scraped from real, uncontrolled field conditions.	A field-condition counterweight to PlantVillage
BRACOL	Arabica coffee leaf disease and pest images.	Directly relevant to Noor's crop.
WFP food prices (via HDX)	Market price series for staples across many countries.	The independent price reference Noor lacks when the buyer arrives.
CHIRPS	Gridded historical rainfall estimates.	Grounds "40 mm overnight" in something real.
iSDAsoil and SoilGrids	Gridded soil property maps: pH, nutrients, texture. iSDAsoil covers Africa at 30 m; SoilGrids is the global equivalent.	Context for why the third row is struggling; free, and citable at plot level.
Digital Earth Africa	Analysis-ready ready Sentinel-2 and Landsat and Landsat satellite imagery over Africa.	Cropland condition and change over time, if your tool uses any earth observation.
Sentinel-2 and Landsat	Open satellite imagery, global coverage, free registration via the Copernicus Data Space.	Cropland mapping and land use anywhere, where Digital Earth Africa stops at the continent.
NASA POWER	Free agro-climate data: rainfall, temperature, solar, by coordinates.	A no-registration way to time an action to the weather that actually happened.
FAOSTAT	Production, yield and price statistics by country and crop.	Background numbers for your problem case.

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Annex C. Tourism

C.1 The scenario

Noor's farm is not marketed on a digital platform. The visitors who find it arrive and have help translating through a local guide, and she has no way of knowing what value she created, what worked during the tourists' visits and what did not once they leave.

- **The problem.** Tourism is a major job creator in the economies where Noor lives, and small operators are where most of that employment sits. Their ability to hire and grow depends on customers they can reach and on understanding the value those customers actually take home. Most small operators run on instinct: they know visitors leave happy, but not why, not which parts of the experience are worth building on, and not how to turn a good review into a returning customer, a referral, or a new product. Weak digital presence and marketing that rarely extends past word of mouth compound the gap.
- **What other digital tools have been in use.** In Jordan, the World Bank has supported work to bring youth and small operators onto digital platforms and into online services, including training operators to list and manage their business on booking and marketing platforms. Getting an operator visible online with a listing, a booking page, digital payments, already unlocks customers without any AI.
- **Where AI can add value.** Small language models can read across a scattered set of reviews and messages and tell the operator what visitors keep returning to, what they wish had been different, and which parts of the experience are worth building the next product around; the analytical work most small operators cannot do for themselves.
- **Pre-conditions to keep in mind.** In Jordan, the binding constraint for informal operators was the absence of a digital listing and the skills to manage one.

> **Hackathon challenge**
>
> Design and demonstrate a Small AI solution that helps Noor, as a small tourism operator, complete one meaningful business workflow; for example, becoming discoverable, communicating across languages, responding to enquiries, shaping an experience, managing a booking, learning from vistor feedback or following up with a guest.

C.2 Datasets, extended

This annex lists the to agriculture datasets only — the second of the two layers described in Section 7.1. Every entry can also draw on the common datasets in Section 7.3, which are open to all three sectors.

Tourism is the sector where the voice and language table below carries the most weight, because the core task is multilingual.

FOR TOURISM		
Dataset	**What it is**	**Why it matters here**

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UN Tourism statistics	Arrivals, receipts and tourism industries by country and year.	Problem-is-real evidence for how much visitor money matters in your named country.
World Development Indicators, tourism series (World Bank)	International tourism receipts, arrivals and expenditure as country indicators.	A second, citable source, and one that connects tourism to jobs and GDP.
World Bank Enterprise Surveys	Firm-level survey data across more than 150 economies, including the smallest firms.	Evidence on what actually constrains a small operator: finance, skills, competition, informality.
OpenStreetMap via Overpass	Open map data: accommodation, attractions, roads, points of interest.	The findability baseline: what a visitor searching near Ondera would and would not discover today.
Wikivoyage	Openly licensed travel guide text in many languages.	Multilingual tourism text for building or testing what visitors ask about.
Yelp Open Dataset	Millions of real business reviews, released for academic use.	Real review language for the "what did visitors actually like" analysis; check the license terms fit your use.
MASSIVE (Amazon)	About one million short utterances in 51 languages, labelled by intent.	The closest open template for sorting visitor requests into your fixed list: a price, directions, a booking.
FLORES-200 / NLLB-200 (Meta)	Translation benchmark and open models across 200 languages.	The message arrives in a language Noor cannot read; this is where the reply draft starts.

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